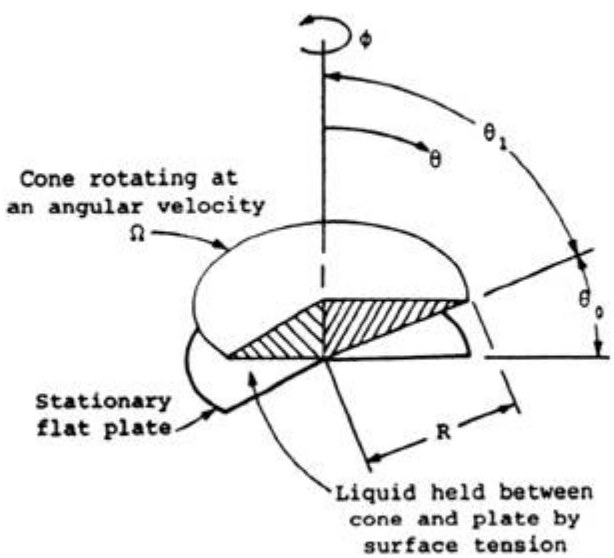
CHE 311 Homework Week 7

Apply continuity equation and Navier-Stokes equation to solve flow problems

1. Using Navier-Stokes function to recalculate the cone-and-plate viscosity problem. Consider the system illustrated in the figure. An inverted cone rests on a stationary flat plate such that its apex just touches the plate. The set – up is placed in a puddle of a sample of the liquid whose viscosity needs to be determined. The cone rotates at an angular velocity Ω. Assuming that the torque required to turn the cone can be determined, develop expressions for the torque $\tau_{\theta\phi}$ as a function of r and θ. Assume only one stress component $\tau_{\theta\phi}$ is significant and that it is constant throughout the fluid. End effects can be eliminated.



2. Two coaxial cylinders with radii R0 and Ri rotate in the same direction but at different angular velocities of ω 0 and ω i respectively. Develop the velocity profile of the fluid filling the space between them. Assume isothermal laminar flow.

3. BSL 3A.1

4. BSL 3A.4